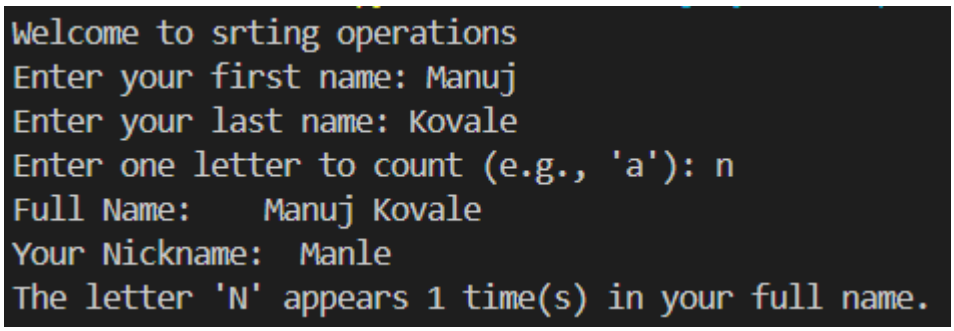
Exp no.:-6

Title:-String Operations

Q1) Write a Python program to read a user's first and last name, create a nickname by slicing and concatenating parts of both names, search for a specific letter in the full name, and display the result using string formatting.

- def analyze_name():

```
print("Welcome to srting operations ")
first_name = input("Enter your first name: ").strip().title()
last_name = input("Enter your last name: ").strip().title()
search_letter = input("Enter one letter to count (e.g., 'a'): ").strip().lower()
full_name = f"{first_name} {last_name}"
nickname = f"{first_name[:3]}{last_name[-2:]}".title()
letter_to_check = search_letter if len(search_letter) == 1 and search_letter.isalpha() else 's'
letter_count = full_name.lower().count(letter_to_check)
print(f"Full Name:    {full_name}")
print(f"Your Nickname:  {nickname}")
print(f"The letter '{letter_to_check.upper()}' appears {letter_count} time(s) in your full name.")
analyze_name()
```



2. Write a program to process the string "PythonProgrammingBasics", extract specific parts using slicing, join them using concatenation, check if a substring exists, and display the formatted result.

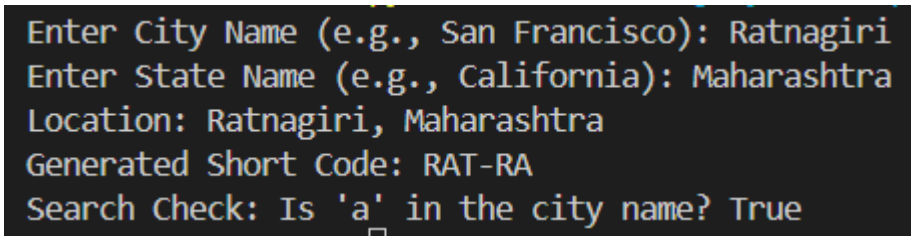
```
def process_string_data():
    data_string = "PythonProgrammingBasics"
    part_1 = data_string[0:6]
    part_2 = data_string[17:]
    new_string = part_1 + " " + part_2 + "!"
    search_term = "Program"
    is_found = search_term in data_string
    print(f"Original String: {data_string}")
    print(f"Extracted 'Python': '{part_1}'")
    print(f"Extracted 'Basics': '{part_2}'")
    print(f"Concatenated New String: '{new_string}'")
    print(f"Check for '{search_term}': {is_found}"
    process_string_data()
```

```
Original String: PythonProgrammingBasics
Extracted 'Python': 'Python'
Extracted 'Basics': 'Basics'
Concatenated New String: 'Python Basics!'
Check for 'Program': True
```

3. Write a program to accept city and state names, generate a short code using slicing and concatenation, search for a character in the city name, and print all information using string formatting.

```
def generate_city_code():
    city = input("Enter City Name (e.g., San Francisco): ").strip().title()
    state = input("Enter State Name (e.g., California): ").strip().title()
    search_char = 'a'
    city_part = city[:3]
```

```
    state_part = state[-2:]
    short_code = city_part.upper() + "-" + state_part.upper()
    is_char_present = search_char in city.lower()
    print(f"Location: {city}, {state}")
    print(f"Generated Short Code: {short_code}")
    print(f"Search Check: Is '{search_char}' in the city name? {is_char_present}")
    generate_city_code()
```



4. Develop a program that reads an email address, extracts the username before '@', concatenates it with a new domain, checks for digits in the username, and displays the formatted new email.

```
def process_email():
    original_email = input("Enter an email address (e.g., user123@oldmail.com): ").strip()
    at_index = original_email.find('@')
    if at_index == -1:
        print("Error: Invalid email format (missing '@').")
        return
    username = original_email[:at_index]
    new_domain = "@new-secure.net"
    new_email = username + new_domain
    has_digit = any(char.isdigit() for char in username)
    print(f"Original Email: {original_email}")
    print(f"Extracted Username: {username}")
    print(f"New Email Address: {new_email}")
    print(f"Does the username contain digits? {has_digit}")
```

process_email()

```
Enter an email address (e.g., user123@oldmail.com): kowalemanuj@gmail.com
Original Email: kowalemanuj@gmail.com
Extracted Username: kowalemanuj
New Email Address: kowalemanuj@new-secure.net
Does the username contain digits? False
```

5. Write a Python program that takes a word, reverses it using slicing, concatenates it with the original, searches for a matching character, and formats the output for display.

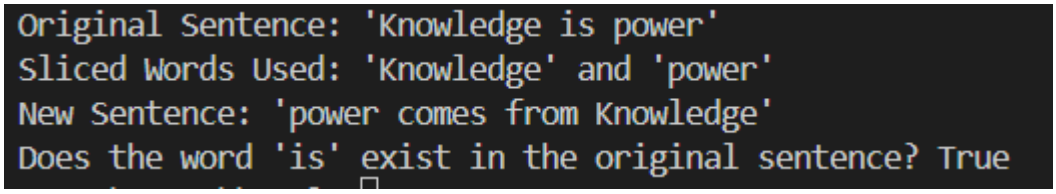
```
def reverse_and_check_word():
    original_word = input("Enter a word: ").strip()
    search_char = 'o'
    reversed_word = original_word[::-1]
    combined_word = original_word + reversed_word
    is_char_found = search_char in combined_word.lower()
    print(f"Original Word: {original_word}")
    print(f"Reversed Word: {reversed_word}")
    print(f"Combined String: {combined_word}")
    print(f"Is the character '{search_char}' present? {is_char_found}")
if __name__ == "__main__":
    reverse_and_check_word()
```

```
Enter a word: Hacker
Original Word: Hacker
Reversed Word: rekcaH
Combined String: HackerrekcaH
Is the character 'o' present? False
```

6. Write a program using the string "Knowledge is power" to extract specific words with slicing, join them to form a new sentence, search for a given word, and print the formatted sentence.

```
def process_fixed_sentence():
    sentence = "Knowledge is power"
```

```
    search_word = "is"
    word_1 = sentence[0:9]
    word_3 = sentence[13:18]
    new_sentence = f"{word_3} comes from {word_1}"
    is_word_found = search_word in sentence
    print(f"Original Sentence: '{sentence}'")
    print(f"Sliced Words Used: '{word_1}' and '{word_3}'")
    print(f"New Sentence: '{new_sentence}'")
    print(f"Does the word '{search_word}' exist in the original sentence? {is_word_found}")
if __name__ == "__main__":
    process_fixed_sentence()
```



7. Write a Python program that accepts two words, extracts specific portions using slicing, concatenates them into a phrase, searches for "ing", and displays the result in a formatted manner.

```
def make_phrase():
    word_a = input("Enter first word (e.g., learn): ").strip()
    word_b = input("Enter second word (e.g., coding): ").strip()
    search_substring = "ing"
    part_a = word_a[:2]
    part_b = word_b[-4:]
    new_phrase = part_a + " is " + part_b
    is_ing_present = search_substring in word_b.lower()
    print(f"Original Words: {word_a}, {word_b}")
    print(f"Generated Phrase: '{new_phrase}'")
```

```
    print(f"Substring '{search_substring}' check in Word B: {is_ing_present}")
if __name__ == "__main__":
    make_phrase()
```

```
Enter first word (e.g., learn): Hacking
Enter second word (e.g., coding): Learning
Original Words: Hacking, Learning
Generated Phrase: 'Ha is ning'
Substring 'ing' check in Word B: True
```

8. Write a program that accepts two sentences, extracts parts from both using slicing, combines them, searches for a common substring, and prints a formatted comparison message.

```
def combine_sentences():
    sentence_1 = input("Enter the first sentence: ").strip()
    sentence_2 = input("Enter the second sentence: ").strip()
    search_text = "the"
    part_1 = sentence_1[:10]
    part_2 = sentence_2[-10:]
    combined_message = f"BEGIN: {part_1} ... {part_2} :END"
    is_found = search_text in combined_message.lower()
    print(f"Part 1 (S1 Start): '{part_1}'")
    print(f"Part 2 (S2 End):   '{part_2}'")
    print(f"Full Combined Message: {combined_message}")
    print(f"Is '{search_text}' in the combined message? {is_found}")
if __name__ == "__main__":
    combine_sentences()
```

```
Enter the first sentence: Welcome to Python
Enter the second sentence: thanks
Part 1 (S1 Start): 'Welcome to'
Part 2 (S2 End):   'thanks'
Full Combined Message: BEGIN: Welcome to ... thanks :END
Is 'the' in the combined message? False
```

9. Write a Python program to accept a movie name and release year, create a short movie code using slicing and concatenation, check for a specific word in the movie name, and format the final output.

```
def generate_movie_code():
    movie_name = input("Enter Movie Name (e.g., The Great Gatsby): ").strip().title()
    release_year = input("Enter Release Year (e.g., 2013): ").strip()
    search_word = 'the'
    movie_part = movie_name[:3]
    year_part = release_year[-2:]
    movie_code = movie_part.upper() + year_part
    is_word_present = search_word in movie_name.lower()
    print(f"Movie: {movie_name} ({release_year})")
    print(f"Generated Movie Code: {movie_code}")
    print(f"Search Check: Does '{search_word}' appear in the title? {is_word_present}")
if __name__ == "__main__":
    generate_movie_code()
```

```
Enter Movie Name (e.g., The Great Gatsby): URI
Enter Release Year (e.g., 2013): 2019
Movie: Uri (2019)
Generated Movie Code: URI19
Search Check: Does 'the' appear in the title? False
```

10. Write a Python program to generate a simple password by slicing and concatenating parts of a user's name and birth year, search for digits in it, and display the password using formatted output.

```
def generate_simple_password():
    user_name = input("Enter your full name (e.g., John Cena): ").strip().replace(" ", "")
    birth_year = input("Enter your Birth Year (e.g., 1995): ").strip()
    mid_start = len(user_name) // 2 - 2
    mid_end = len(user_name) // 2 + 2
    name_part = user_name[mid_start:mid_end]
    year_part = birth_year[:2]
    simple_password = name_part.lower() + year_part + "!"
    has_digits = any(char.isdigit() for char in simple_password)
    print(f"Generated Password: {simple_password}")
    print(f"Password Length: {len(simple_password)}")
    print(f"Does the password contain digits? {has_digits}")
if __name__ == "__main__":
    generate_simple_password()
```

```
Enter your full name (e.g., John Cena): Manuj
Enter your Birth Year (e.g., 1995): 2004
Generated Password: manu20!
Password Length: 7
Does the password contain digits? True
```

11. Write a program that accepts a college name and department, creates a department code using slicing and concatenation, searches for the word "Tech", and prints the formatted details.

```
def generate_dept_code():
    college_name = input("Enter College Name (e.g., Massachusetts Institute): ").strip()
    department = input("Enter Department Name (e.g., Electrical Engineering): ").strip()
    search_word = "Tech"
    college_char = college_name[0]
    dept_part = department[:4]
    dept_code = college_char.upper() + "-" + dept_part.upper()
```

```
    is_tech_present = search_word.lower() in college_name.lower()
    print(f"College: {college_name}")
    print(f"Department: {department}")
    print(f"Generated Department Code: {dept_code}")
    print(f"Search Check: Is '{search_word}' in the college name? {is_tech_present}")
if __name__ == "__main__":
    generate_dept_code()
```

```
Enter College Name (e.g., Massachusetts Institute): SGU
Enter Department Name (e.g., Electrical Engineering): CSE
College: SGU
Department: CSE
Generated Department Code: S-CSE
Search Check: Is 'Tech' in the college name? False
```

13. Write a Python program that reads a product name and category, forms a label using sliced parts, searches for a specific substring, and prints the final label using string formatting.

```
def create_product_label():
    product_name = input("Enter Product Name (e.g., Ultra-Speed Mousepad): ").strip()
    category = input("Enter Category (e.g., Gaming Peripherals): ").strip()
    search_substring = "Pad"
    prod_part = product_name[:5]
    cat_part = category[-3:]
    product_label = f"SKU-{prod_part.upper()}{cat_part.upper()}"
    is_substring_present = search_substring.lower() in product_name.lower()
    print(f"Product: {product_name}")
    print(f"Category: {category}")
    print(f"Product Label: {product_label}")
    print(f"Search Check: Is '{search_substring}' in the product name?
{is_substring_present}")
```

```
if __name__ == "__main__":
    create_product_label()
```

```
Enter Product Name (e.g., Ultra-Speed Mousepad): Mouse
Enter Category (e.g., Gaming Peripherals): Gaming
Product: Mouse
Category: Gaming
Product Label: SKU-MOUSEING
Search Check: Is 'Pad' in the product name? False
```

14. Write a program to input a book title and author name, slice and join parts of both, search for a word in the title, and display the information in a single formatted string.

```
def display_book_info():
    book_title = input("Enter Book Title (e.g., The Art of War): ").strip()
    author_name = input("Enter Author Name (e.g., Sun Tzu): ").strip()
    search_word = "Art"
    title_part = book_title[:4]
    author_part = author_name[-4:]
    summary_snippet = f"Title Start: '{title_part}', Author End: '{author_part}'"
    is_word_found = search_word in book_title
    print(f"Full Title: {book_title}")
    print(f"Snippet: {summary_snippet}")
    print(f"Title Check: Is the word '{search_word}' in the title? {is_word_found}")
if __name__ == "__main__":
    display_book_info()
```

```
Enter Book Title (e.g., The Art of War): Art gallery
Enter Author Name (e.g., Sun Tzu): Joseph
Full Title: Art gallery
Snippet: Title Start: 'Art ', Author End: 'seph'
Title Check: Is the word 'Art' in the title? True
```

15. Write a Python program to create a social media username by slicing and concatenating parts of a user's name and year of birth, search for vowels in it, and display the formatted username.

```
def generate_social_username():
    full_name = input("Enter your full name (e.g., Alice Wonderland): ").strip().replace(" ", "")
    birth_year = input("Enter your Birth Year : ").strip()
    vowels = "aeiou"
    vowel_count = 0
    name_part = full_name[:5].lower()
    year_part = birth_year[-4:]
    username = name_part + "_" + year_part
    for char in username:
        if char in vowels:
            vowel_count += 1
    print(f"Generated Username: @{username}")
    print(f"Username Length: {len(username)}")
    print(f"Number of Vowels Found: {vowel_count}")
if __name__ == "__main__":
    generate_social_username()
```

```
Enter your full name (e.g., Alice Wonderland): Manuj Kovale
Enter your Birth Year : 2004
Generated Username: @manuj_2004
Username Length: 10
Number of Vowels Found: 2
```